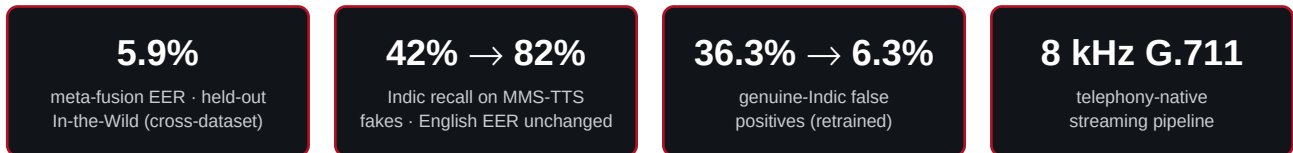


VoxShield

Real-Time Voice-Deepfake Detection for Indic-Language Telephony Fraud

An on-premises, Indic-first, telephony-hardened prototype that detects AI-cloned voices in phone-quality Indian-language calls and returns a calibrated, explainable fraud-risk verdict within the first seconds of a call (streaming).



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1 Executive Summary

Problem. AI voice cloning has made convincing synthetic speech cheap and fast, and it is increasingly used in phone-based financial scams in India — often in Indian languages and over ordinary, low-quality telephone lines. **Why now.** Public authorities in India have flagged a sharp rise in digital financial fraud and in AI-enabled impersonation scams [1][2]. **Gap.** Most voice-deepfake detectors are tuned for clean, English-centric, cloud-processed audio; their effectiveness on Indian-language, narrowband (8 kHz) telephone audio and inside an institution's own infrastructure varies and is rarely reported.

What VoxShield is. A real-time detector purpose-built for this setting: it ingests G.711 μ -law 8 kHz telephony audio, runs five complementary detectors under a meta-fusion layer with Indic- and telephony-aware routing, and outputs a LOW/MEDIUM/HIGH risk verdict with explainable reason codes to drive a step-up-verification workflow — not automatic blocking. **Evidence.** On a held-out In-the-Wild set (cross-dataset), the learned meta-fusion reaches **5.9% EER** — versus 16.0% for the best single detector and 19.1% for naive averaging, so the observed gain comes from the learned fusion, not any single model. After Indic-aware retraining, recall on MMS-TTS Indic deepfakes rose **42% → 82%** with English EER unchanged at 5.9%, and genuine-Indic false positives fell **36.3% → 6.3%**. Scores are Platt-calibrated (ECE 0.044). These are research-corpus and generator-/language-specific evaluations, not production metrics.

2 The Problem

Three trends intersect. First, modern text-to-speech and voice-conversion systems can produce natural-sounding speech from short reference samples, lowering the cost and skill needed to impersonate a voice. Second, a large share of Indian fraud is delivered over the phone — impersonation of bank officials, "digital arrest" intimidation, and relative-in-distress scams [3] — including documented cases of AI voice cloning used against Indian targets [14]. Third, much of this happens in Indian languages and over degraded telephone channels.



Figure 1 — The compounding risk chain that defines VoxShield's operating context.

For a fraud team, the operational problem is not an academic 'is this audio synthetic?' label. It is: can we flag a suspicious live call early enough, in the right language, over a real phone line, with a reason a human analyst can act on — without moving sensitive customer audio out of controlled infrastructure?

3 Why Existing Detection Breaks in This Setting

The difficulty is a set of domain gaps between where detectors are usually built and where Indian telephony fraud actually occurs:

Dimension	Common assumption	This setting
Audio channel	Studio / wideband	8 kHz G.711 narrowband, codec loss, noise
Language	English-centric data	Indic phonetics; code-switching
Attack coverage	Single detector / known generators	Diverse, evolving TTS & voice-conversion
Output	Black-box score	Operational risk + reason for an analyst
Deployment	Cloud inference	Controlled / on-prem infrastructure

Narrowband telephony discards exactly the high-frequency detail (roughly above 3.4 kHz) where many synthesis artifacts live [4], so a model trained on clean speech can degrade on phone audio. Indic-language coverage in public anti-spoofing data is limited, so an English-tuned detector can both miss Indic fakes and over-flag genuine Indic speakers — the two failure modes VoxShield targets directly in §8.

4 The VoxShield Solution

In one line: VoxShield listens to a call the way a phone network delivers it, asks five complementary detector families ('is this voice real?'), and combines them with a learned, calibrated meta-stacker into a single risk score plus a plain-language reason — routed through paths specialised for Indian languages and telephone audio.

- **Indic-first** — an Indic-aware detection path with language-ID routing, not English models applied as an afterthought.
- **Telephony-hardened** — built and evaluated on G.711 μ -law 8 kHz narrowband conditions.
- **Learned multi-detector fusion** — five complementary detector families combined by a trained stacker, so no single model is a point of failure.
- **Explainable, calibrated risk** — Platt-calibrated LOW/MEDIUM/HIGH with reason codes, not a bare binary label.
- **Liveness option** — a random challenge-response provides an additional signal against replayed/pre-recorded audio (implemented; not yet experimentally validated).
- **Honesty engineered** — never auto-blocks; every verdict carries reason codes and a SHA-256 audit record.
- **On-prem architecture** — designed so raw call audio can stay inside the institution's environment.
- **Fraud-workflow native** — detect → explain → risk-score → trigger step-up verification.

5 System Architecture

VoxShield is a streaming pipeline. Audio arrives in telephony format, is gated for speech, sliced into overlapping windows, and scored by parallel detectors whose outputs are fused and routed to a calibrated risk decision and reason codes.

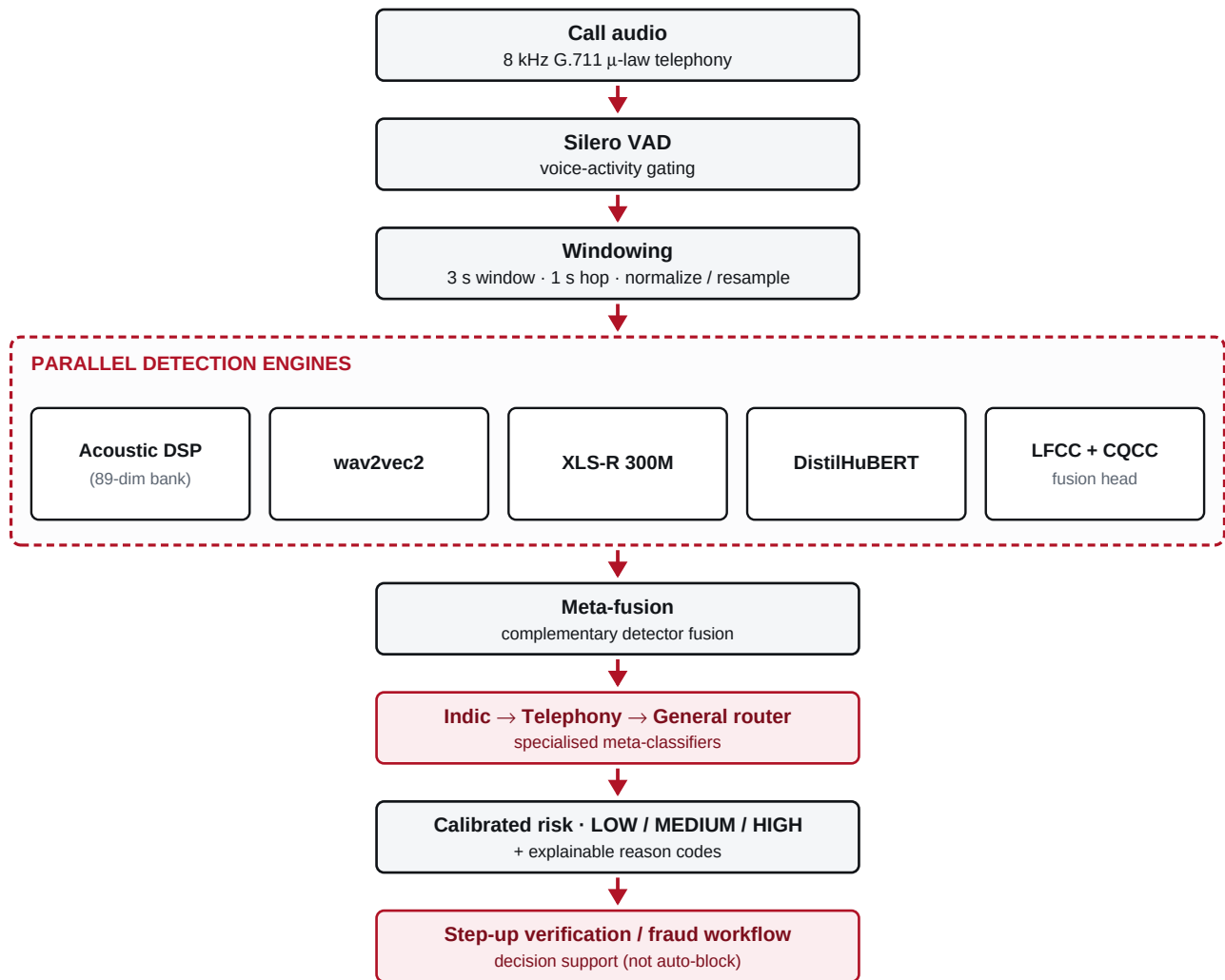


Figure 2 — VoxShield streaming architecture, from telephony audio to fraud-workflow trigger.

6 Model Architecture & Multi-Detector Fusion

VoxShield deliberately mixes detector families on the hypothesis that they respond differently to synthesis artifacts and channel distortions, so fusion may improve robustness. It combines self-supervised speech representations — wav2vec2 [5], wav2vec2-XLS-R-53 (multilingual, 300M) [6], and DistilHuBERT [11] — with classical spoofing-sensitive acoustic features (LFCC and CQCC [12]) and a DSP detector over an 89-dimensional feature bank (LFCC, CQCC, group-delay phase, F0/prosody, jitter, shimmer, and a high-frequency >6 kHz band). Features are computed at a 16 kHz analysis rate (8 kHz Nyquist), so the >6 kHz band carries information only for wideband audio; under 8 kHz G.711 telephony it is essentially empty, and VoxShield uses that emptiness as a narrowband indicator rather than relying on high-frequency artifacts there. A meta-classifier fuses the detector outputs; a router then prioritises the Indic-aware meta-detector, then the telephony/narrowband meta-detector, then the general detector.

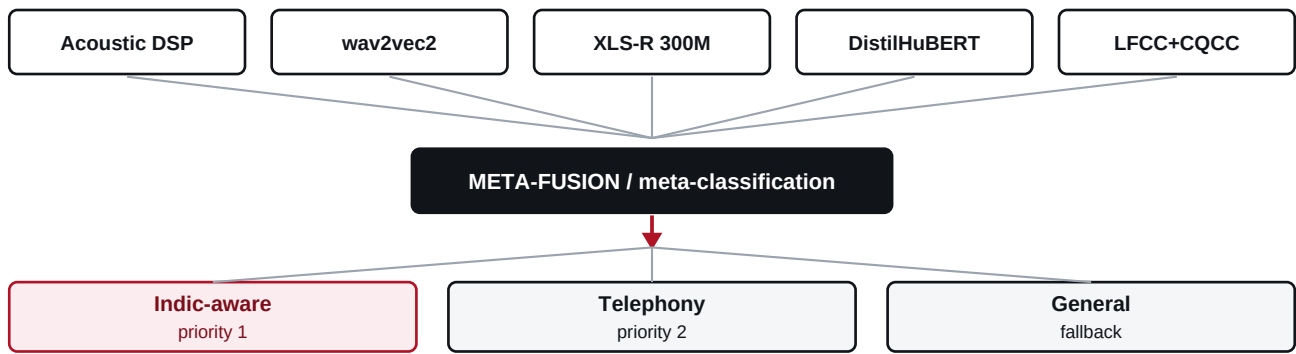


Figure 3 — Complementary detectors feed a meta-fusion layer with Indic → telephony → general routing.

7 Methodology

Inference pipeline. Call audio (8 kHz G.711 μ -law) → Silero VAD [10] → 3-second windows at a 1-second hop → an 89-dimensional acoustic feature vector computed at a 16 kHz analysis rate (LFCC, CQCC, group-delay phase, F0/prosody, jitter, shimmer, and a >6 kHz high-frequency band that is informative for wideband audio and empty under 8 kHz telephony) → five detectors scored in parallel → a learned logistic meta-stacker → Platt calibration → a LOW/MEDIUM/HIGH verdict with reason codes. Windows are re-scored continuously so a verdict forms early and is refined as the call proceeds.

Evaluation protocol. The headline detection number is a *cross-dataset* figure: detectors are trained on other corpora and evaluated on held-out In-the-Wild [7] audio the models did not see in training — a deliberately hard generalisation test rather than an in-distribution score. Per-detector and fusion scores are computed on a common scored evaluation set so the stacker can be compared directly against each single model and against naive averaging. EER (Equal Error Rate) is the operating point where false-accept and false-reject rates are equal [8]; calibration quality is reported with Expected Calibration Error (ECE) and Brier score. Indic evaluation uses genuine Indic speech for false positives and MMS-TTS-generated [9] Indic speech for fake recall; these are generator- and language-specific and are reported separately from the English detection metric.

Experiment	Data / generator	Train→Test	Metric	Result	Scope
Detection	In-the-Wild [7]	other corpora → held-out ITW	EER	5.9%	cross-dataset; speaker overlap not documented
Detector comparison	common scored set	—	EER	16.0–62.4% single; 19.1% naive; 5.9% learned	same eval set
Indic fake recall	MMS-TTS [9] Indic	before/after Indic retrain	Recall	42%→82%	single generator (MMS-TTS)
Genuine-Indic FP	genuine Indic speech	channel-aware	FPR	36.3%→6.3%	tested Indic conditions
Calibration	held-out	—	ECE / Brier	0.044 / 0.048	calibration distribution

What is measured vs engineered. Detection EER, the detector/fusion comparison, calibration, and the Indic recall / false-positive figures are **measured**. Telephony robustness is primarily **engineered** via G.711/8 kHz augmentation and a narrowband path; per-window compute latency and live-network robustness are **not yet formally benchmarked** and are flagged as such throughout.

8 Telephony Hardening

Telephone audio is not studio audio. VoxShield's pipeline is native to G.711 μ -law at 8 kHz, with the ~300–3400 Hz narrowband limit [4], codec distortion, background noise and packet loss treated as first-class conditions rather than nuisances. Speech is gated with Silero VAD [10] and processed in 3-second sliding windows at a 1-second hop, so a verdict can form within the first seconds of a call and be refined as more

audio arrives.

Stage	Studio speech	Telephone speech	VoxShield handling
Bandwidth	Full-band ($\leq \sim 20$ kHz)	$\sim 0.3\text{--}3.4$ kHz	narrowband-aware features + routing
Sample rate	16–48 kHz	8 kHz	native 8 kHz path
Coding	Lossless / high-rate	G.711 μ -law	codec-condition training/eval
Artifacts kept	HF synthesis cues	much HF removed	DSP + fusion recover residual cues

What is actually tested vs engineered — an important distinction for telephony claims:

Telephony condition	Status in this work
G.711 μ -law 8 kHz inference pipeline	Implemented
Narrowband / codec training augmentation	Implemented
Simulated narrowband (augmented) evaluation	Evaluated (augmented conditions)
Real PSTN / SIP captured-call evaluation	Pending (not yet measured)

9 Indic-Aware Detection

A multilingual pretrained backbone does **not** automatically make a deepfake detector language-robust: synthesis artifacts and genuine-speaker distributions both shift across languages, so a detector tuned on English can misjudge Indic speech in both directions — missing Indic fakes and over-flagging genuine Indic speakers. Public Indian-language resources such as AI4Bharat’s IndicVoices [13] help ground genuine Indic speech, while multilingual TTS such as Meta’s MMS [9] can synthesise Indic fakes. VoxShield adds an Indic-aware detection path with on-device language-ID routing and retrains on Indic material. Measured on the tested Indic conditions, this moved two metrics substantially — **without regressing English**, whose EER stayed at 5.9%:

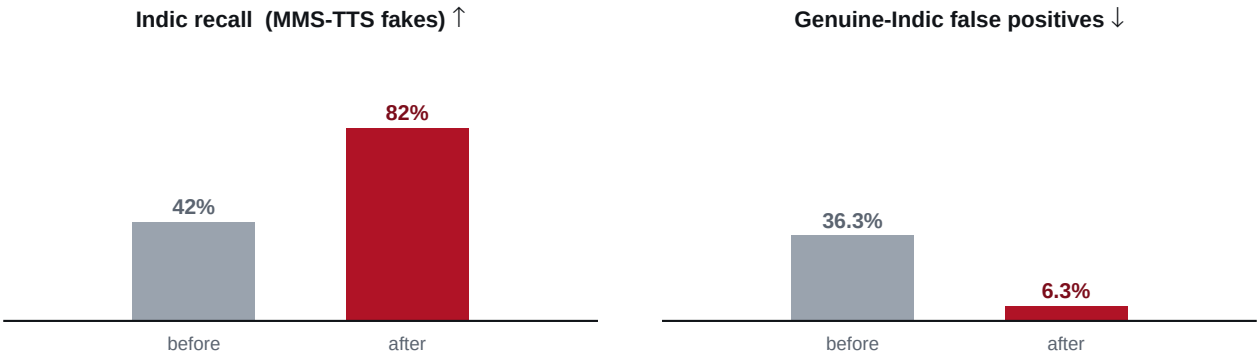


Figure 4 — Indic-aware retraining: recall on MMS-TTS [9] Indic deepfakes and false-positive rate on genuine Indic speech. These figures are specific to the tested Indic conditions and should not be read as covering every Indian language.

The false-positive result is the one a bank cares about most: over-flagging genuine customers is expensive and erodes trust, so cutting genuine-Indic false positives from 36.3% to 6.3% is as important as raising recall.

10 Results & Ablation

Every number is labelled with its exact evaluation. EER is not accuracy; Indic recall is not overall recall; a false-positive rate is not a false-negative rate. The headline results:

Evaluation setting	Metric	Baseline	VoxShield	Scope / reading
Held-out In-the-Wild [7] (cross-dataset)	EER	16.0%*	5.9%	*best single model; fusion nearly 3× better

MMS-TTS [9] Indic deepfakes	Recall	42%	82%	generator-specific; English EER unchanged (5.9%)
Genuine Indic speech	False-positive rate	36.3%	6.3%	channel-aware; fewer genuine-user flags
Held-out In-the-Wild	ROC-AUC	—	0.983	threshold-independent separability

Why multi-detector fusion? (ablation). We measured the standard single-model approaches first. On the same evaluation, individual detectors range from 16.0% to 62.4% EER, and — critically — **naive averaging (19.1%) is worse than the best single model (16.0%)**. Only the *learned* meta-stacker, which assigns each detector a weight (some negative, i.e. it learns which detectors to distrust), reaches 5.9%. Under this evaluation the evidence supports the value of *learned* fusion: the observed performance gain (EER) comes from the fusion, not any single model. This does not by itself establish universal fusion superiority.

Equal Error Rate by configuration (lower is better)

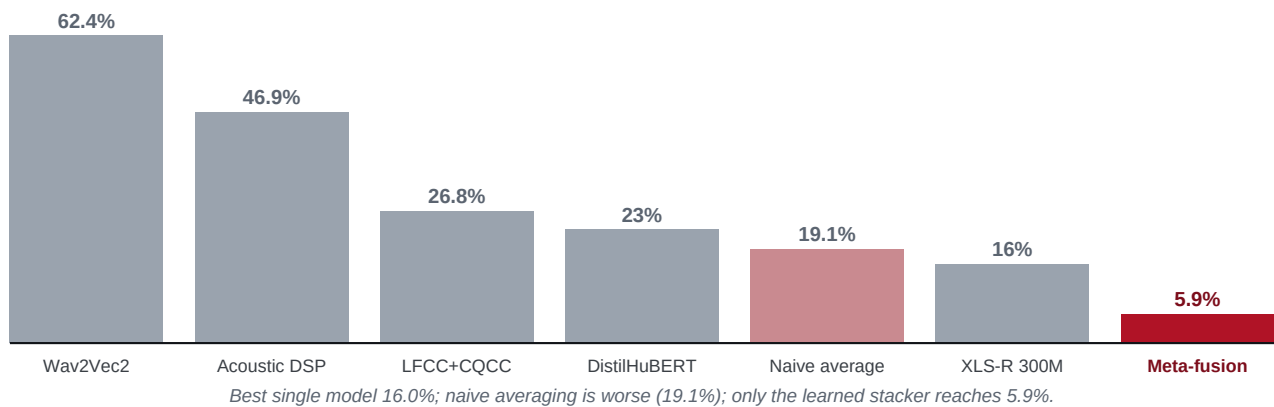


Figure 5 — Equal Error Rate by configuration. Fusion beats every single detector and, importantly, beats naive averaging — evidence that the gain is from learned weighting, not model choice alone.

Calibration. After Platt calibration, scores are intended to approximate empirical event probabilities under the calibration distribution — what a bank acts on rather than a raw ranking; calibration quality is **ECE 0.044**, **Brier 0.048**, and an abstain band leaves the uncertain middle unforced. Probabilistic validity outside the evaluated distribution is not claimed. **Scope.** All figures are research-corpus / generator-/language-specific evaluations under held-out (and, for telephony, augmented narrowband) conditions — not production or live-network measurements. VoxShield is a working prototype (TRL-5), not a deployed service.

11 Real-Time Operation & Explainability

Streaming, not batch. Scoring runs on 3-second windows at a 1-second hop, so a first verdict forms as soon as a 3-second window fills and is then refined continuously — a verdict after the call ends is forensics, not protection. This is a streaming-capable architecture designed for early-call risk formation; **per-window compute latency and end-to-end system latency are not yet formally benchmarked** on target hardware and are stated as pending, not claimed.

Explainability. Each verdict carries reason codes derived from the detector outputs and the 89-dim feature bank, so an analyst sees *why* a call scored high rather than a bare label. The implemented codes are:

- **SSL** — high neural-synthesis likelihood from the self-supervised detectors
- **PH** — unnaturally regular phase / group-delay structure
- **HF** — anomalous energy in the >6 kHz band (wideband audio only; not available under 8 kHz telephony)
- **PR** — missing or flattened prosody (F0, jitter, shimmer)
- **BR** — absence of natural breath (TTS inserts silence, not breathing)

These are signal-derived indicators surfaced to the analyst; we do not claim they are formally faithfulness-tested explanations.

12 Fraud Workflow Integration

VoxShield is a decision-support layer, not an autonomous account-blocking system. Its output feeds an institution's existing fraud process:

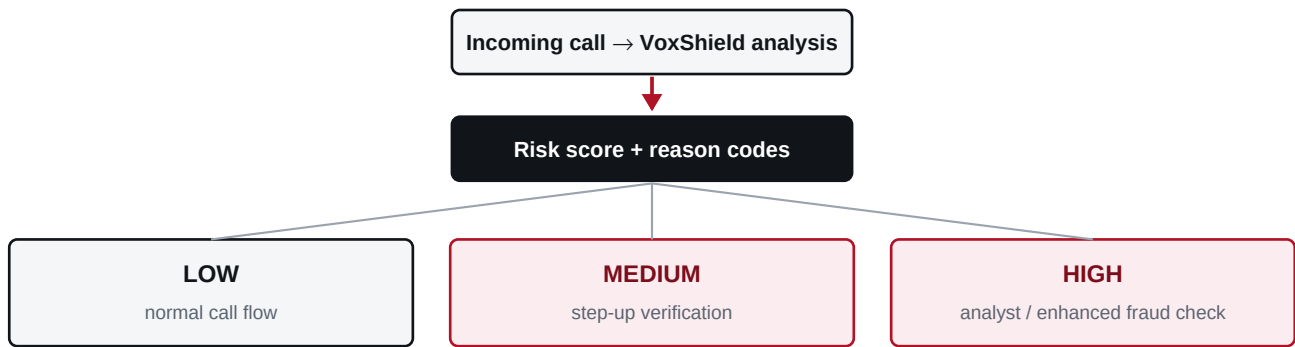


Figure 6 — Risk-tiered routing into an existing fraud workflow; humans stay in the loop.

13 Security & Deployment

VoxShield is architected to support institutional data-governance requirements: it can run on controlled or on-premises infrastructure, so raw call audio need not be sent to an external inference service, and it can operate in air-gapped configurations where required. The design anticipates auditability, model/version management, and secure integration with banking and contact-centre systems. We do not claim formal regulatory certification; a compliance assessment against specific frameworks would be part of any real deployment.

14 Innovation & Defensibility

The contribution is not a single new algorithm — the component techniques are published. It is the **integration under one hard set of constraints**: Indic-language adaptation + telephony-native processing + learned multi-detector fusion + specialised routing + calibrated explainable risk + fraud-workflow integration + controlled-infrastructure deployment, holding all of them simultaneously. Each constraint individually is known; satisfying them together — e.g. keeping English EER flat while lifting Indic recall, on 8 kHz audio, on-prem — is the engineering difficulty.

Implemented today (measured)	Future defensibility (proposed)
Indic-aware path + language-ID routing	Privacy-preserving institutional data flywheel
G.711 8 kHz telephony hardening	Proprietary Indic synthetic-voice corpus
Learned five-detector meta-stacker + calibration	Broader generator diversity & continual eval
Explainable reason codes + SHA-256 audit	Bank-specific calibration
Liveness challenge-response option	Production-scale SIP/VoIP integration
On-prem / air-gapped architecture	Federated cross-institution learning

The left column is built and measured; the right column is future work and is labelled as such throughout.

15 Scalability

The same pipeline is intended to sit behind bank contact centres, telco fraud systems and SIP/VoIP gateways, feeding fraud-analyst dashboards. Streaming, windowed inference makes per-call cost predictable; on-prem deployment keeps data in place. Concrete throughput and latency figures are deferred to measured benchmarks on target hardware (future work).

16 Limitations & Validation Targets

We state these plainly; several are active validation targets rather than solved problems.

- **Cross-generator generalisation** — the Indic fake evidence uses MMS-TTS as the generator; generator-disjoint testing (train on some synthesis systems, test on entirely unseen ones) is a key validation target and is not yet demonstrated.

- **Speaker disjointness** — train/test speaker overlap is not documented here; speaker-disjoint evaluation is needed to isolate spoof detection from speaker-memorisation effects.
- **Real-network telephony** — telephony results are on augmented narrowband, not captured PSTN/SIP audio; real-network evaluation is pending.
- **Very short segments** and heavy background noise reduce reliability.
- **Code-switching and language imbalance** — Indic results are specific to tested conditions, not all Indian languages.
- **Adversarial adaptation** — attackers aware of the detector may adapt.
- **Benchmark-to-production gap** — corpus/augmented results are not live-network guarantees.

17 Future Work

Measured latency/throughput benchmarks on target hardware; a formal ablation report; more Indic languages and a larger real-world fraud corpus; privacy-preserving federated learning across institutions; continuous model adaptation and better calibration; adversarial-robustness hardening; and measured production-scale SIP integration with human-in-the-loop fraud operations.

18 Conclusion

VoxShield addresses a specific, under-served deployment gap: Indic-language + telephony-quality + real-time + controlled-infrastructure + explainable voice-deepfake detection for financial fraud. It is an honest prototype with measured, carefully-scoped evidence — 5.9% EER on held-out In-the-Wild audio, and large Indic-specific gains in recall and false-positive rate after Indic-aware retraining — and a clear separation between what is built and what is proposed.

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Public artifacts: GitHub (DG10911/voxshield) · HuggingFace (dg10911/voxshield-checkpoints) · Team DigiSeva · Contact: devanshgoenka03@gmail.com